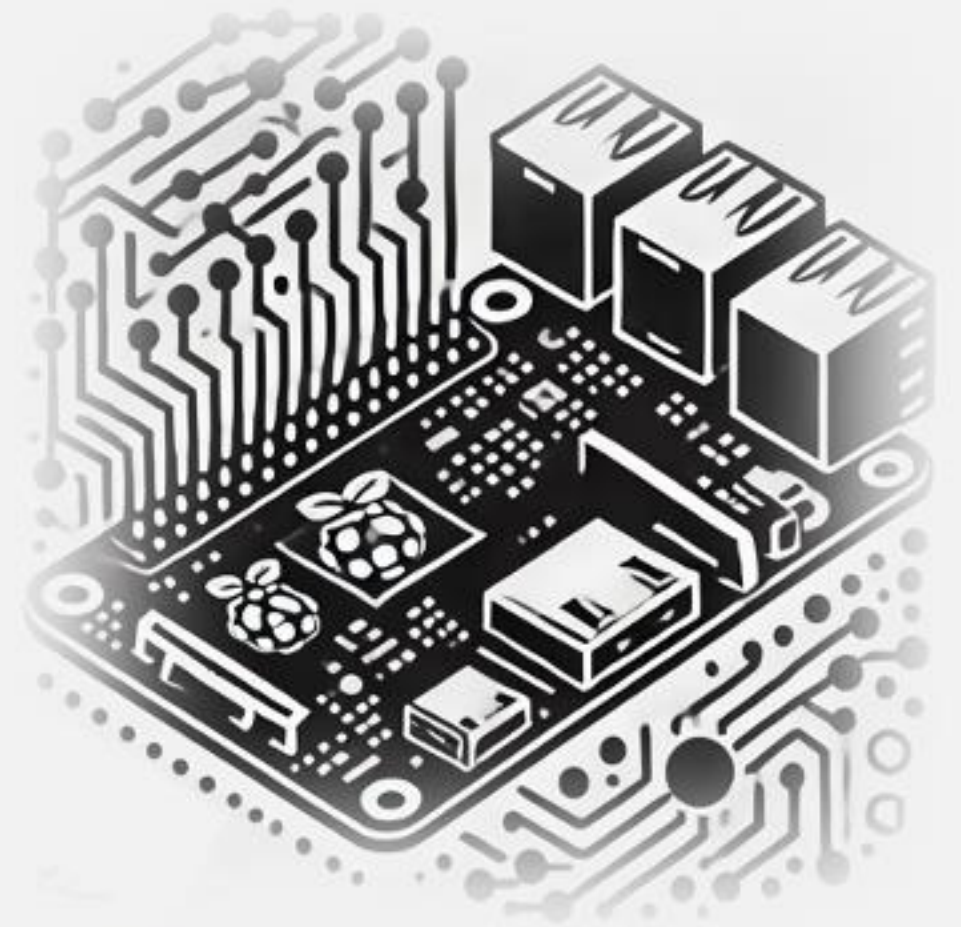


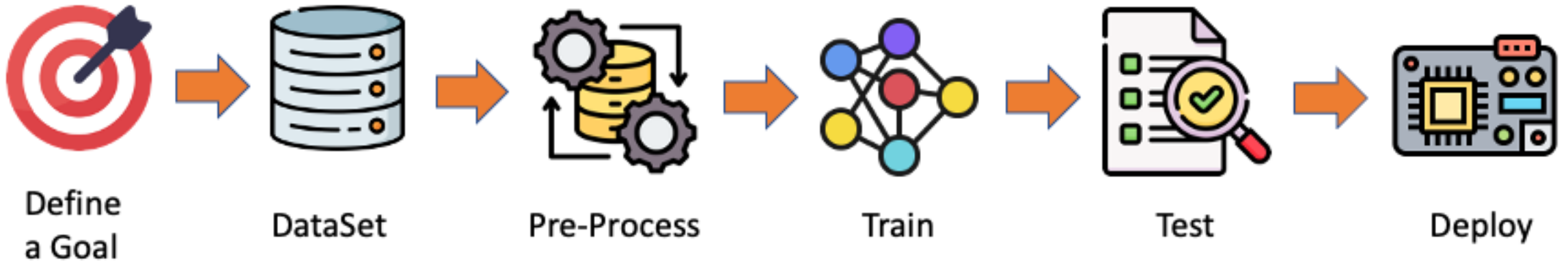
IESTI05 – Edge AI

Machine Learning System Engineering

6. Image Classification Project: Goal and Data Collection



Machine Learning Workflow



Machine Learning Workflow

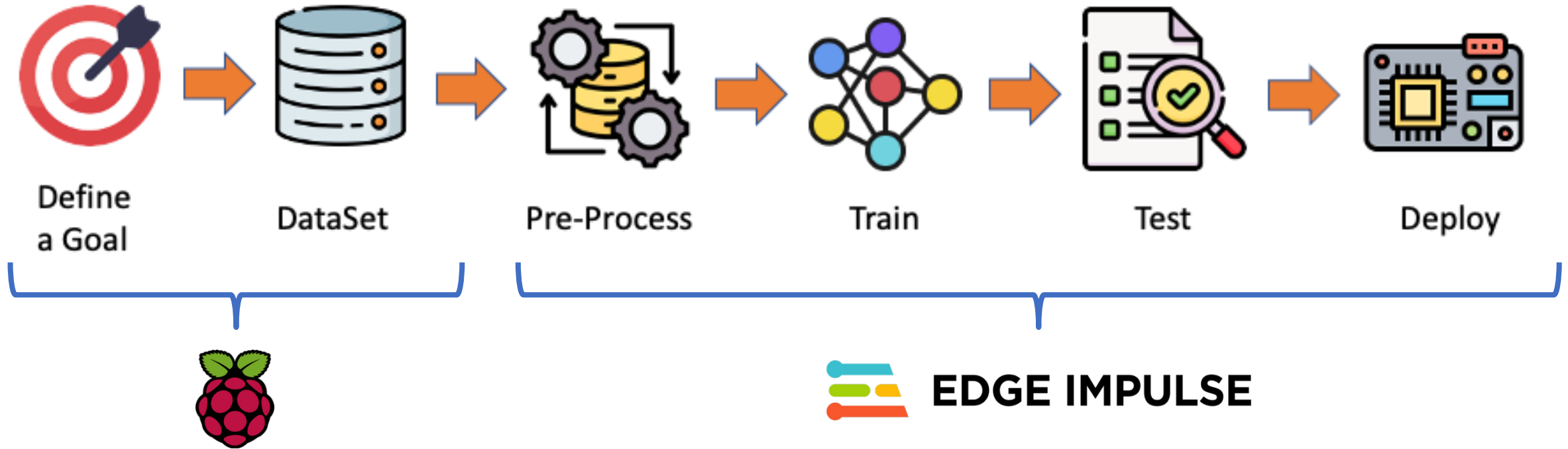


Image Classification Example



Define
a Goal

- Classes:
 - medicine
 - background



<https://studio.edgeimpulse.com/public/114253/latest>

Image Classification Example



Define
a Goal

- Classes:
 - mug
 - background



<https://studio.edgeimpulse.com/public/139479/latest>

Img Class Example

Medicine Classifier

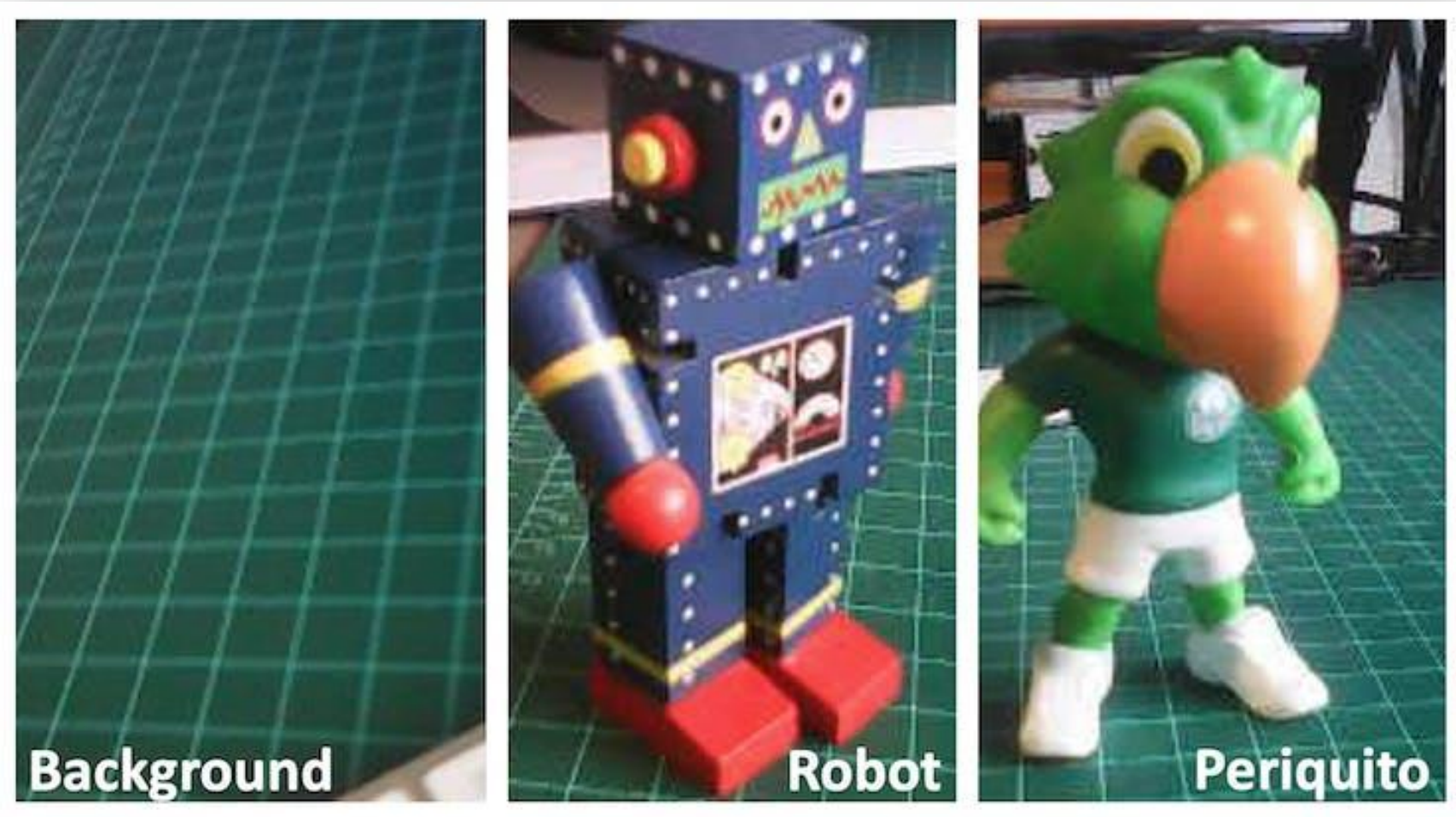


Define
a Goal

- Classes:
 - prodR
 - prodD
 - prodT
 - backG



The Goal



Data Collection Guidelines



Set up a **web server** on the Raspberry Pi to view captured images



Use **Flask**, a lightweight web framework for Python



Capture **QVGA (320 x 240)** images with the Raspberry Pi camera



Collect around **60 images** from each category (periquito, robot, and background)

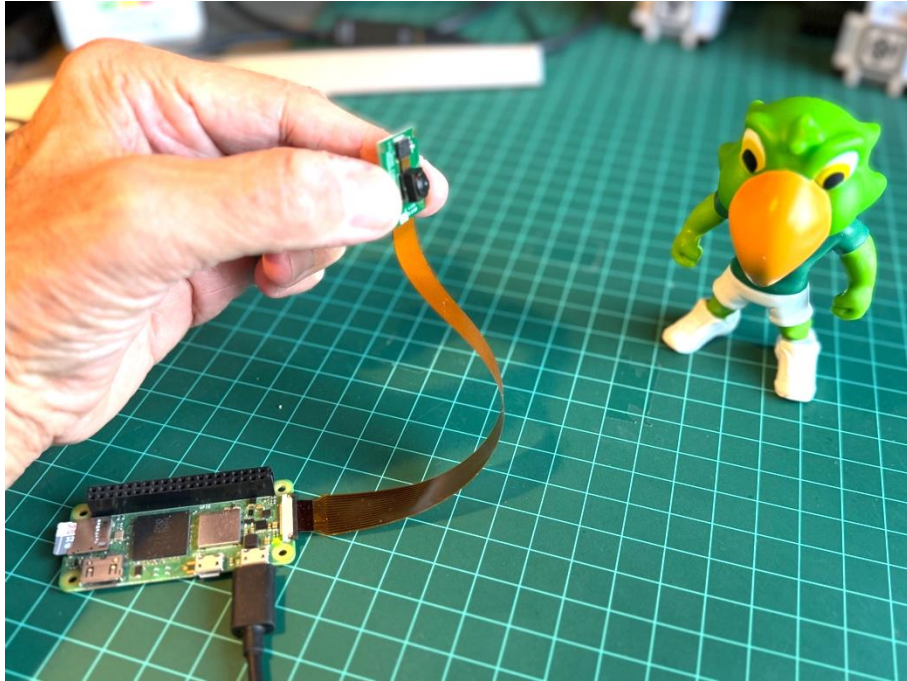


Capture different **angles, backgrounds, and light conditions**



Dataset quality directly impacts model performance

Data Collection – Web App



Web Interface

Accessible from any device on the same network



Live Camera Preview

Shows real-time feed from the camera



Labeling System

Input labels for different image categories



Organized Storage

Saves images in label-specific subdirectories



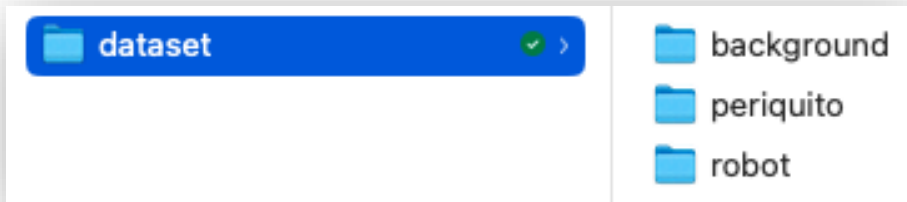
Per-Label Counters

Tracks captured images for each label



Summary Statistics

Provides summary when stopping capture



Setting up a **Venv**, installing **Flask**, and running the **App**

Activate the environment:

```
source ~/tflite_env/bin/activate
```

Install Flask

```
pip3 install flask
```

Run the app

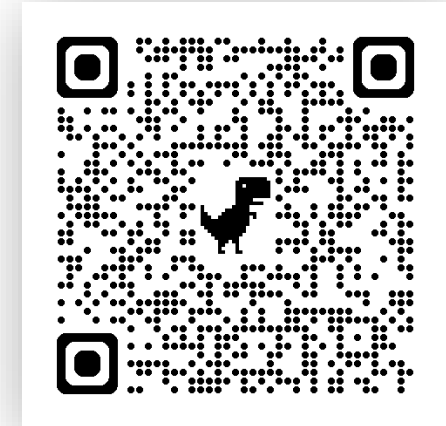
```
python get_img_data.py
```

To **exit** the virtual environment, use:

```
deactivate
```

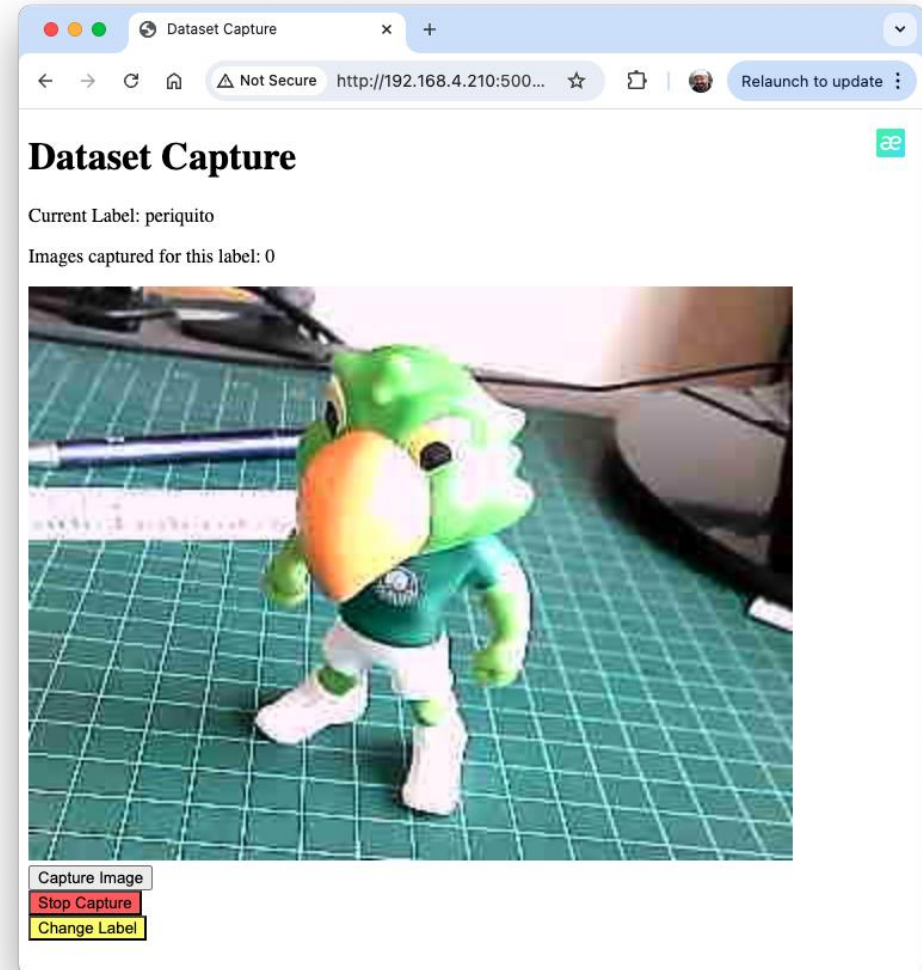
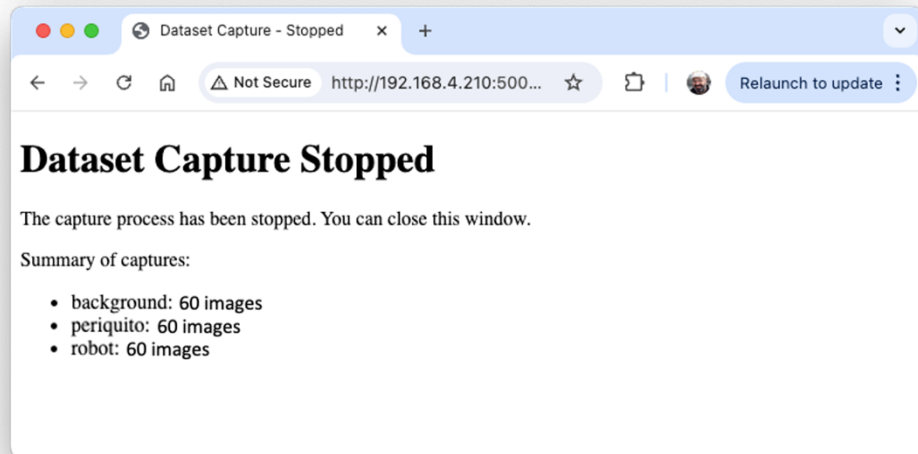
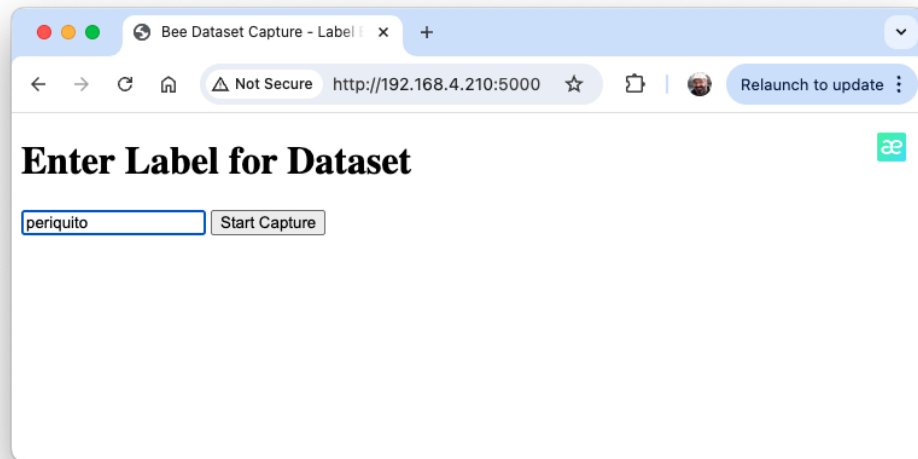


Flask



[get_img_data.py](#)

The Web App



Questions?



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